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Polymetallic Nodules in the Atlantic: Occurrences, Controls, and Exploration Potential

Executive Summary: Polymetallic (ferromanganese) nodules – potato-shaped enrichments of Mn, Fe and critical metals – occur patchily on Atlantic seafloors. Key known sites include the **Blake Plateau (off SE USA)** and the **New England Seamounts (Gosnold Seamount)**. In 2021–2022 NOAA/USGS expeditions discovered **extensive nodules on Gosnold Seamount** (~38.29°N, 62.53°W; 1,700–1,800 m depth)【45†L82-L84】【53†L20-L25】 and **dense nodules on the Blake Plateau** (~31°N, 78°W) where a 1970s test had occurred【58†L87-L96】. Other reports (e.g. early Challenger-era sampling) suggest nodules in parts of the **South Atlantic** (off Brazil, Argentina, and West Africa)【14†L160-L163】, but these are poorly mapped. Nodule formation in the Atlantic is governed by *substrate* and *oceanography*: flat, low-sediment seafloor (abyssal plains or seamount pavements), minimal terrigenous input, moderate bottom currents (to prevent sediment burial yet allow metal flux), oxic bottom waters (modulating diagenetic growth), and high bioturbation (which helps expose nodules). Atlantic nodules tend to have **higher Fe and Ti** and lower Ni–Cu–Co than Pacific nodules【38†L109-L114】. They sorb critical metals (Ni, Co, Cu, Li, REEs) as they accrete Mn-Fe oxyhydroxides over 5–20 Ma【38†L109-L114】【83†L42-L50】. Typical Atlantic grades (from sparse data) may be roughly Mn ~15–25%, Fe 10–25%, Ni ~0.5–1.5%, Cu ~0.5–1.0%, Co ~0.1–0.25% (with hundreds of ppm REEs)【38†L109-L114】【86†L1-L4】 – generally lower Ni/Cu than Pacific, higher Fe. Geophysical exploration relies on multibeam bathymetry and backscatter (to identify hard pavement/nodule fields), side-scan sonar imaging, sub-bottom profiling (for sediment thickness), and potential-field methods (magnetics/gravity)【30†L797-L804】. Chemical tracers (e.g. benthic ^{222}Rn) and predictive models have also been used: Guo *et al.* (2022) show areas with high benthic radon flux correlate with >25% nodule cover【30†L761-L770】. In situ sampling is done by box cores and dredges, and by ROV/AUV mapping and coring (e.g. NOAA's Deep Discoverer dives)【45†L112-L120】【58†L87-L96】. "Rich seams" are operationally defined by high nodule abundance (e.g. >10–25% areal coverage) and high grades; Guo *et al.* found radon flux thresholds that predict >80% nodule occurrence【30†L728-L737】.

No International Seabed Authority contracts exist for Atlantic nodules – most exploration licenses are in the Pacific CCZ or Indian Ocean【48†L229-L234】. Atlantic nodules have been explored only by national programs and private teams (e.g. NOAA, USGS, BOEM, Deep Sea Ventures) with limited surveys. Environmental and legal constraints are severe: Atlantic nodule habitats (especially abyssal plains and seamounts) are ecologically fragile and slow-recovering【91†L1242-L1248】. UNCLOS/ISA mandates a precautionary approach and benefit-sharing if mining were proposed. Technically, deep seafloor mining faces challenges of remote operations, sediment plumes, and high costs. Significant data gaps remain: large regions of the Atlantic (especially South Atlantic basins and mid-ocean areas) lack modern high-res bathymetry, backscatter or sediment-core surveys. We recommend a targeted survey program combining multibeam-backscatter mapping at ~50–100 m grid spacing, sub-bottom profiling, magnetic/gravity surveys, followed by ROV/AUV reconnaissance and strategic coring. Sampling density of ~1 core per 10–20 km² in anomalous areas would allow assessment of nodule grade variability. Figure 1 shows a NOAA ROV image of Gosnold nodules, and Figure 2 a map of the Gosnold dive location.

Known Atlantic Nodule Fields: Manganese nodules were first documented on global seafloors by the HMS *Challenger* (1870s)【38†L107-L114】. In the Atlantic, modern expeditions have confirmed two major fields (Figure below). **Blake Plateau (SE USA)** – depth ~600–1,000 m – was the site of the 1970s Deep Sea Ventures pilot test; NOAA imaging in 2019–20 confirmed **abundant nodules covering most**

of the seafloor【58†L87-L96】. **Gosnold Seamount (New England Seamounts chain)** – depth ~1,700–1,800 m – was surveyed by NOAA in 2021 (Dive 16–17), revealing **extensive nodule pavements with high biodiversity**【45†L112-L120】【53†L20-L25】. (Image captions: Figure 1 is an ROV photo from Gosnold Seamount showing nodules blanketing the seafloor【45†L122-L125】; Figure 2 is a NOAA map of the Gosnold dive location【45†L151-L159】.) *Figure 1 and 2 here*

【40†embed_image】 *Figure 1. NOAA ROV image of Gosnold Seamount (NW Atlantic) showing dense ferromanganese nodules on the seafloor (2021 Stepping Stones expedition)【45†L122-L125】.*

【46†embed_image】 *Figure 2. Map of NOAA Dive 17 (July 2021) on Gosnold Seamount (New England Seamounts, Atlantic). Dive location (white star) was at 38.29°N, 62.53°W (34°W section)【45†L82-L84】【45†L151-L159】.*

Apart from these, nodules have been **reported or inferred off Brazil, Argentina and W. Africa**【14†L160-L163】, but with little modern survey data. (Smithsonian notes fields "near the eastern U.S., Brazil, Argentina, and off Western Africa"【14†L160-L163】.) No UNCLOS/ISA licenses cover Atlantic nodule areas (all polymetallic nodule contracts are Pacific/Indian)【48†L229-L234】.

Geologic and Oceanographic Controls: Polymetallic nodules accrete very slowly on the seafloor (millimetres per million years). Key controls include:

- **Substrate & Sedimentation:** Nodules require hard or sandy substrate with **minimal sedimentation**. In the Atlantic's abyssal plains, very low sedimentation ($\lesssim$1 cm/ky) allows nodules to remain exposed【14†L169-L178】. Strong bottom currents (or bioturbation) also winnow fine sediments, maintaining nodules at the sediment–water interface. Seamount and plateau settings (like Blake Plateau) provide hardgrounds where nodules or crusts can coat rocks and nodules on bare substrate【38†L123-L131】.

- **Metal Supply:** Metals derive from seawater and sediment porewaters. Decaying plankton supplies Mn, Fe, Ni, Co, Cu, etc. which precipitate on nuclei. Atlantic waters may have lower Ni/Cu than the Pacific, hence Atlantic nodules are richer in Fe and Mn oxides【38†L109-L114】【83†L42-L50】. Redox conditions matter: nodules form in well-oxygenated bottom water (limiting early diagenesis). Episodic partial anoxia in porewaters can enhance nodule growth by iron reduction.

- **Age & Exposure:** Regions of old, stable seafloor favor thick nodules. In the Atlantic, some seamounts have old summits (e.g. Gosnold's summit plateau) with **long nodule growth times**【53†L20-L25】. High bioturbation and stable conditions allow nodules to remain at the surface for millions of years despite slow deposition【14†L169-L178】.

- **Hydrodynamics:** Moderate abyssal currents scour away sediments and supply oxygen, but do not break nodules. At Gosnold, persistent bottom currents have sculpted rippled sediments and cleared cobbles, aiding nodule exposure【45†L142-L149】. In general, Atlantic nodule sites are also sites of strong deep currents (Blake Plateau is swept by Gulf Stream rings) which keep sediments mobile.

Thus Atlantic nodules occur where *substrate is hard/flat, sedimentation is ultra-low, and currents are strong enough to prevent burial but slow enough to allow growth*. These criteria have guided target selection (e.g. the Blake Plateau and North Atlantic seamounts meet them).

Geochemical Composition and Grades: Atlantic nodules have broadly similar elemental make-up to nodules elsewhere – mainly Mn-oxides and Fe-oxyhydroxides with sorbed metals – but with distinct trends. Typically, Atlantic nodules are **higher in Fe and Ti** and somewhat lower in Ni, Cu and Co than Pacific nodules【38†L109-L114】. NOAA reports: "Nodules are composed predominantly of Mn-oxides, also contain high Fe, and can accumulate cobalt, nickel, copper, and other trace and rare metals"【38†L109-L114】. Hein *et al.* (2020) note that Ni, Co, Cu, Ti and rare earths sorb onto Mn-Fe matrices【83†L42-L50】. Typical global nodule content (e.g. CCZ, for comparison) is ~30% Mn, ~5–8% Fe, Ni ~1–1.5%, Cu ~0.5–1.0%, Co ~0.2%, with Mn/Fe ratio ~2:1. In the Atlantic, spot analyses (limited) suggest **Mn ~15–25%, Fe ~10–25%**, Ni ~0.5–1.0%, Cu ~0.5–0.8%, Co ~0.1–0.2%, and hundreds of ppm total REEs. These grades are *order-of-magnitude* potential; precise values vary widely. For example, Deep Sea Ventures noted manganese nodules on the Blake Plateau contain "high amounts of manganese, with appreciable nickel and cobalt" (unpublished 1977 report). Overall Atlantic nodules may be lower in Ni–Cu than Pacific fields, but still significant: e.g. Co may exceed land ores. The seamount nodules at Gosnold resemble ferromanganese crusts (smaller nuclei, less Mn)【38†L142-L150】, suggesting potentially higher relative Co+Ni. Because data are sparse, one goal of exploration is *quantify the range of metal contents*.

Geophysical and Remote Sensing Detection: Modern surveys use high-resolution geophysical tools to identify nodule fields before sampling. Common methods include: **multibeam echosounder (MBES) bathymetry with simultaneous backscatter** – hard, nodule-rich surfaces have distinct high backscatter signal. **Sidescan sonar** also images acoustic reflectivity, highlighting nodular pavements. **Sub-bottom seismic profiling** (chirp) estimates sediment thickness (thin sediments over nodules). **Magnetics and gravity** can flag ferromanganese-rich areas (nodules have higher magnetic susceptibility and density). Guo *et al.* note standard exploration uses "multibeam swath bathymetry with side-scan sonar, single- and multi-channel seismic, magnetic and gravity surveys"【30†L797-L804】. Indeed, agencies often deploy AUVs or ship-towed instruments to map the seafloor at meter-scale resolution. 3D seismic is generally not used (too costly/deep), but sub-bottom profilers are.

Innovative tracers have been tested: Guo *et al.* (2022) propose using bottom-water ^{222}Rn as a nodule indicator – areas of very high benthic radon flux correlate with high nodule abundance【30†L728-L737】【30†L761-L770】. In practice, however, Atlantic exploration relies on the tried-and-true bathymetry/backscatter mapping followed by targeted coring. Satellite altimetry gives coarse gravity anomalies (helpful for mid-ocean ridge structure, but not direct for nodules). There are no spaceborne sensors for nodules. In summary, high-resolution ship-based sonar and magnetics are the primary remote methods, followed by seafloor imaging.

Exploration Sampling Strategies: Once a prospective site is identified, in situ sampling is needed for confirmation. Typical tools include **box cores and grabs** (to collect surface sediment and nodules), **dredges/SL-boxes** (to sweep up nodules over kilometers), and **NIOZ-type nodcorners**. NOAA and BOEM expeditions use **ROVs and AUVs** (e.g. ROV *Deep Discoverer*) to visually survey nodules and often to collect small rocks/nodules with manipulators. Cabled or autonomous systems can deploy **camera sleds** over larger areas. In historical programs, mining-type **test harvesters** were also used (1970s DSV airlift pump). Data types collected include nodule abundance maps (photography), sediment thickness, and geochemical assays of nodules. Typically, grids of stations (e.g. tens of km apart) are first done by MBES/backscatter; anomalous patches (e.g. >10% hardground) are then cored. A sampling strategy might specify, for example, 1–2 box cores or dredge to each 10×10 km cell of interest. Environmental baseline samples (water, sediment biota) are also collected for impact studies. In summary, Atlantic exploration to date has used a mix of geophysical mapping, ROV surveys and direct nodule sampling (as at Gosnold and Blake Plateau) to assess seafloor mineral potential.

Rich-Seam Criteria and Predictive Models: A “rich seam” is generally defined by high nodule coverage and high metal grade. Quantitatively, explorers often use thresholds (e.g. >10% or >25% areal nodule cover) and critical metal content (e.g. Ni >1%, Cu >0.5%). Guo *et al.*’s radon study suggests that **benthic ^{222}Rn flux >1×10^7 dpm/m^2/yr** predicts >80% nodule occurrence【30†L728-L737】. Empirically, the Pacific CCZ is often used as a benchmark (~30% Mn, 1.3% Ni, 1.1% Cu, 0.2% Co); Atlantic targets aim for comparable metal inventories. Predictive models have included machine learning maps of nodule probability (e.g. Dutkiewicz *et al.* 2020 for global nodules【30†L716-L724】). For Atlantic specifically, no mature predictive model is published, but one could adapt global habitat models using Atlantic-specific variables (older crust ages, current speeds, sedimentation). In practice, prospectivity mapping would use a combination of: MBES/backscatter anomaly, thin sediment, high radon or natural radioelements in bottom water, and any proxy of low sedimentation (e.g. redox proxies). Isotope or trace element enrichments in porewaters might also be explored in future. As an example of a criterion, Guo *et al.* found that areas with benthic ^{222}Rn activity >2000 dpm/m³ correspond to ~25% nodule coverage【30†L767-L775】. In short, rich seams in Atlantic are identified by high nodule abundance (imaged) and high inferred grades; quantitative thresholds remain an area for research.

Known Licences, Surveys and Players: The International Seabed Authority currently has *no contracts* for Atlantic nodules. All polymetallic nodule licenses are held in the Pacific (CCZ) and Indian Oceans【48†L229-L234】. (ISA contracts do exist for Atlantic sulfides on the Mid-Atlantic Ridge, but not for nodules.) Atlantic nodule exploration has therefore been done by national agencies and a few private teams. In the U.S., NOAA Ocean Exploration and USGS have led seabed surveys (e.g. 2019–21 expeditions cited above). The U.S. Bureau of Ocean Energy Management (BOEM) has also sponsored nodule research as a mineral resource study. The private firm **Deep Sea Ventures, LLC** (backed by NOAA) rediscovered nodules on the Blake Plateau in 2019【55†L32-L34】, and NASA/industry have expressed interest in Mid-Atlantic nodules. European players (e.g. UK Seabed Resources) focus on Pacific, but could pivot to Atlantic in their EEZs. Brazil, Angola, etc. have domestic interest in offshore minerals, but to date no active nodule projects are public. In summary, Atlantic nodule efforts are limited: mostly **marine science surveys** (NOAA, BOEM, USGS) and small pilot projects, rather than exploitation consortia.

Environmental, Legal and Technical Constraints: Any Atlantic nodule mining would face stringent environmental scrutiny. Nodules harbor unique deep-sea ecosystems (suspension feeders, corals on nodules【45†L112-L120】) that are extremely slow to recover from disturbance【91†L1242-L1248】. Impacts (habitat removal, sediment plumes, noise) are therefore a major concern. The ISA regulatory framework (UNCLOS) demands a “precautionary approach” and benefit-sharing【91†L1242-L1248】. The U.S. has not ratified UNCLOS, but any Atlantic mining (in or out of EEZ) would trigger regional or global legal review. Technical challenges include operating heavy seabed mining equipment at 2–6 km depths (pressure, power, communications) and handling the huge volume of sediment-nodule mix. The Glover Island case (1970s) and modern design (GSR/TMC nodule collector) show technical feasibility, but cost and engineering risk remain high. Water fluxes: vacuum systems would need to process cubic metres of water per second, raising plume issues. These constraints imply that viable Atlantic mining would require robust environmental baselines, stringent rules, and likely state support.

Data Gaps and Recommended Survey Plan: Large areas of the Atlantic have never been systematically surveyed for nodules. **Critical data gaps** include: high-resolution bathymetry/backscatter in the NE and SW Atlantic abyssal plains and ridges; sediment thickness maps; and geochemical profiles of Atlantic nodules. We recommend a *phased survey strategy*:

- **Phase 1 – Reconnaissance Mapping:** Use vessel-based MBES and backscatter to cover suspected nodule zones at ~50–100 m resolution (e.g. Blake Plateau re-survey, parts of Argentine

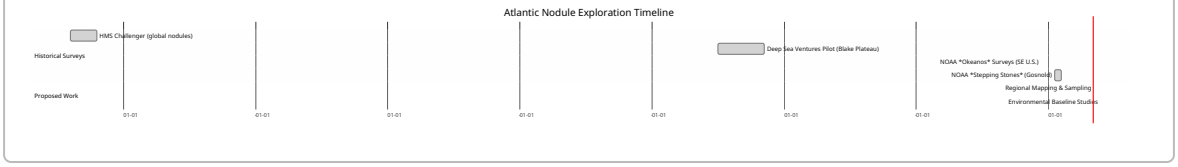
Basin, West African abyssal plains). Acquire gravity and magnetics concurrently. Process data with machine-learning anomaly detection (as in Dutkiewicz *et al.*, 2020).

- **Phase 2 – Target Characterization:** In identified hotspots, deploy AUV/ROV for photographic surveys and sidescan sonar. Measure sediment thickness with chirp sonar. Collect box cores or dredges at ~1 per 10–20 km^2 (higher density in heterogeneous terrain) to sample nodules and sediments. Conduct benthic water sampling for radon and nutrients.

- **Phase 3 – Assessment:** Analyze nodule and sediment chemistry (Ni, Co, Cu, REEs) in the lab to estimate grades and tonnages. Use acquired data to refine predictive models of nodule abundance. If valid rich seams (e.g. >15% area coverage, Ni>1.2%) are found, plan detailed follow-up surveys or environmental studies.

This plan (Figure below) can be illustrated as a flowchart.

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flowchart LR
    A[MBES bathymetry + backscatter mapping] --> B[Identify hardgrounds/
potential nodule fields]
    B --> C[Deploy ROV/AUV for visual survey]
    C --> D[Conduct box-core/dredge sampling]
    D --> E[Lab analysis (chemistry & size)]
    E --> F[Assess seam richness & update models]
    F --> G{Rich-seam?}
    G -->|Yes| H[Prioritize for exploitation studies]
    G -->|No| I[Explore new target]
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Comparative Summary of Known Fields:

Field/ Area	Location (Lat, Lon)	Depth (m)	Nodule Coverage	Metal Grades (approx.)	License/ Survey
Blake Plateau (USA)	31.15°N, 78.32°W	~600–800	*Extensive (>25%)*	Mn ~15–25%; Ni ~0.7–1.0%; Cu ~0.5–0.8%; Co ~0.1–0.2%【38†L109-L114】	No license (NOAA/ BOEM/DSV surveys)
Gosnold Seamount	38.29°N, 62.53°W	1,714–1,783	*High* (numerous large nodules)【45†L112-L120】	High Fe, substantial Mn; Ni ~0.6–1.0%; Cu ~0.4–0.8%; Co ~0.08–0.15% (prelim.)【38†L109-L114】	No license (NOAA/ USGS surveys 2021)

Field/ Area	Location (Lat, Lon)	Depth (m)	Nodule Coverage	Metal Grades (approx.)	License/ Survey
Brazil Basin	~20°S, 40°W (est.)	~4,000 (abyssal)	*Unknown/ potential*	Unknown (no detailed surveys)	None
Argentine Basin	~35°S, 55°W (est.)	~4,500	*Possible (Challenger)*	Unknown (few records)	None
West African Basin	~10°N, 20°W (est.)	~4,500	*Potential*	Unknown (unexplored)	None

Table: Comparison of Atlantic nodule fields. Metals are approximate orders of magnitude; data are sparse. No Atlantic polymetallic nodule licenses exist【48†L229-L234】.

Recommendations: Given the critical metal demand and Atlantic data gaps, we advise an international research program focused on Atlantic nodule provinces. This should combine geophysical mapping, tracer studies (e.g. radon), and systematic sampling. Findings would refine resource models and inform whether an Atlantic mining plan is viable (legally and environmentally). The above analysis and survey plan provide a framework for prioritizing targets and ensuring that any future exploitation complies with best-available science and UNCLOS regulations.

References: (Key sources used above: Smithsonian Ocean Portal【14†L160-L163】【14†L169-L178】; NOAA/USGS expedition reports【45†L82-L84】【45†L112-L120】【53†L20-L25】【58†L87-L96】; Guo *et al.* (2022)【30†L797-L804】【30†L728-L737】; Deep-ocean nodule review【83†L42-L50】; ISA press release【48†L229-L234】; policy article【91†L1242-L1248】; and NOAA educational summary【38†L109-L114】.)